Distributed Spatio-Temporal Processing of Massive Climate Grids for Fire Weather Index Estimation: State-of-the-Art and Architecture Design

Agustín Pizarro, Eduardo Valenzuela, Benjamín Zúñiga
School of Informatics and Telecommunications
Universidad Diego Portales (UDP)
Santiago, Chile
Email: {agustin.pizarro, eduardo.valenzuela, benjamin.zuniga}@mail.udp.cl

Nicolás Hidalgo (Instructor)
School of Informatics and Telecommunications
Universidad Diego Portales (UDP)
Santiago, Chile
Email: nicolas.hidalgoc@mail.udp.cl

***Abstract*—Wildfire danger in Chile has reached catastrophic levels due to a 15-year uninterrupted megadrought and extreme synoptic conditions adhering to the "30-30-30" rule. Operational fire danger assessment relies on the Canadian Forest Fire Weather Index (FWI) System, which is historically evaluated exclusively at sparse, physically vulnerable automatic weather stations (AWS), leaving hazardous spatial blind spots across complex Andean terrain. While high-resolution gridded climate products such as CR2MET v2.5 (∼5 km) provide wall-to-wall spatial coverage, audit of official archives reveals they supply only daily precipitation and temperature extremes, entirely lacking relative humidity and surface wind. To achieve high-resolution, decadal FWI tracking across Chile, a multi-source distributed data pipeline is required to couple CR2MET with hourly ERA5-Land reanalysis and the 35-model NASA NEX-GDDP-CMIP6 climate projection ensemble.**

Crucially, while the arithmetic evaluation of the recursive FWI formulas requires negligible CPU time per cell, the system-level challenge constitutes an authentic distributed systems bottleneck. We formalize this challenge through three distributed dimensions: (i) the *Distributed Corner-Turn / Rechunking Shuffle Problem*, wherein source data stored in time-sliced spatial rasters (`[time, lat, lon]`) must be transposed across a cluster into cell-major time-series (`[cell_id, time]`) to satisfy a 44-year non-commutative, stateful temporal recursion; (ii) multi-resolution spatio-temporal alignment across heterogeneous grids (0.05°, 0.1°, and 0.25°); and (iii) long-term Markovian state persistence and checkpointing.

In this paper, we conduct a rigorous, verified state-of-the-art review covering 18 foundational and contemporary works across distributed processing engines, scientific array frameworks, and fire climatology. We provide a taxonomic comparison demonstrating that existing array systems fail to address multi-decadal, time-recursive stateful scans. Finally, we design a 4-stage distributed architecture using Apache Spark over columnar Apache Parquet with zero-shuffle spatial-first partitioning, evaluated against a Dask/Zarr baseline in a resource-constrained containerized Docker cluster.

***Index Terms*—Distributed Computing, Big Data Systems, Apache Spark, Columnar Storage, Fire Weather Index, Climate Tensors, Rechunking Shuffle, CR2MET, ERA5-Land.**

I. Introduction and Problem Formalization

A. Context and Motivation

During the last decade, Central and Southern Chile have faced devastating wildfire megaseasons, most notably the 2017 "Tormenta de Fuego" ($> 570,000$ ha burned) and the 2023 disaster ($> 430,000$ ha, 26 fatalities) [1], [2]. These catastrophes stem from the interplay of a 15-year uninterrupted megadrought [3], extensive contiguous monoculture plantations of flammable pine and eucalyptus, and recurrent heatwaves governed by the empirical "30-30-30" rule (temperature $T > 30°$C, relative humidity $RH < 30\%$, and 10 m wind speed $W > 30$ km/h).

Operational fire management agencies, including the Chilean National Forestry Corporation (CONAF) and the National Disaster Prevention and Response Service (SENAPRED), rely on the Canadian Forest Fire Danger Rating System (CFFDRS) and its core subsystem, the Fire Weather Index (FWI) [4]. The FWI models the physical drying and moisture dynamics across three distinct forest floor organic layers:

1) **Fine Fuel Moisture Code (FFMC):** Surface litter and cured fine fuels (fast response: 16–24 h timelag).
2) **Duff Moisture Code (DMC):** Moderately compacted decomposing organic material (∼12-day timelag).
3) **Drought Code (DC):** Deep, compacted organic soil layers, serving as a seasonal indicator of cumulative megadrought (∼52-day timelag).

These fuel moisture codes drive three intermediate fire behavior indices: the Initial Spread Index (ISI), Buildup Index (BUI), and the composite Fire Weather Index (FWI) representing frontal flame intensity [5].

B. The Spatial Monitoring Bottleneck and CR2MET Shortfall

Historically, FWI is evaluated exclusively at physical Automated Weather Stations (AWS) operated by the Chilean Weather Service (DMC) and the Water Directorate (DGA). However, Chile's rugged Andean topography creates steep microclimatic gradients across narrow longitudinal spans. The

AWS network is sparse, predominantly urban or coastal, and subject to sensor destruction during wildfires. Consequently, critical rural forest interfaces remain unmonitored "blind spots" where ignitions propagate undetected.

To overcome AWS sparsity, climate researchers developed gridded products such as **CR2MET v2.5** [6], which models daily climate over continental Chile on a regular $0.05^\circ \times 0.05^\circ$ grid ($\sim$5 km resolution, comprising $\sim$20,000 active land cells from 1960 to 2021).

However, an audit of the official CR2MET v2.5 distribution (Zenodo record `10.5281/zenodo.7529682`) reveals a critical operational shortfall: **CR2MET v2.5 exclusively provides daily precipitation (`pr`) and extreme temperatures (`tmin`, `tmax`)**. It does *not* provide relative humidity (RH) or wind speed (W), which are mandatory terms in the differential drying equations of FFMC and ISI.

Therefore, constructing a continuous wall-to-wall FWI database over Chile is impossible using CR2MET alone. It strictly requires a **multi-source distributed data fusion pipeline** that couples CR2MET's 0.05° topographic precipitation/temperature with hourly atmospheric moisture (T_d) and vector wind components (u_{10}, v_{10}) from ECMWF's **ERA5-Land** reanalysis (0.1° grid, 1950–present) [7], and projects historical risk forward across the 35 General Circulation Models (GCMs) and 4 Shared Socioeconomic Pathways (SSPs) of the **NASA NEX-GDDP-CMIP6** ensemble (0.25° grid, 1950–2100) [8].

C. The Distributed Systems Challenge: Why is FWI Genuine Big Data?

At first glance, evaluating the empirical equations of the Canadian FWI might appear computationally trivial. For a single grid cell on a single day, the algebraic formulas for FFMC, DMC, and DC require fewer than 60 floating-point operations (FLOPs). Consequently, evaluating 3.2×10^8 cell-days on a modern CPU executing at several GFLOPs would theoretically take only a few seconds if all data resided in contiguous L1/L2 cache.

The true challenge is fundamentally an **I/O, memory hierarchy, and distributed network communication bottleneck**, characterized by three core system properties:

1) The Distributed Corner-Turn (Rechunking Shuffle) Problem: Let $\mathcal{S}_{c,t} = \langle FFMC_{c,t}, DMC_{c,t}, DC_{c,t} \rangle \in \mathbb{R}^3$ denote the state vector of cell $c \in \{1, \dots, N_c\}$ at day $t \in \{1, \dots, T\}$, and let $\mathbf{x}_{c,t} = \langle T_{c,t}, RH_{c,t}, W_{c,t}, P_{c,t} \rangle \in \mathbb{R}^4$ be the input forcing. The state transition is governed by:

$$\mathcal{S}_{c,t} = \Phi\left(\mathcal{S}_{c,t-1},\, \mathbf{x}_{c,t},\, \text{month}(t),\, \text{lat}(c)\right) \tag{1}$$

Because the physical drying functions inside Φ are exponential, piecewise-linear, and conditioned on rainfall saturation thresholds, Φ is **strictly non-commutative and non-associative across time**:

$$\Phi(\Phi(\mathcal{S}, \mathbf{x}_1), \mathbf{x}_2) \neq \Phi(\Phi(\mathcal{S}, \mathbf{x}_2), \mathbf{x}_1) \tag{2}$$

Thus, time cannot be folded via a parallel tree reduction ($O(\log T)$). The temporal trajectory must be evaluated sequentially as a prefix scan of length T for each spatial cell c.

Crucially, raw climate datasets (NetCDF-4/GRIB2) are archived in **time-major order** (`[time, lat, lon]`), where each 2D file represents a spatial map at time t. If a single node attempts to read the 44-year time series of cell c, it must perform $T \approx 16,000$ non-contiguous, fragmented disk seeks across thousands of compressed NetCDF chunk boundaries. This produces a *Read Amplification Factor* exceeding $10^3\times$, collapsing throughput.

Conversely, if P distributed worker nodes ingest time-sliced blocks in parallel, no single worker possesses the contiguous history of any spatial cell. Grouping observations by cell requires an **all-to-all distributed transpose (corner-turn shuffle)** that transfers:

$$V_{\text{shuffle}} \approx \left(1 - \frac{1}{P}\right) \cdot N_c \cdot T \cdot |\mathbf{x}| \tag{3}$$

across cluster interconnects. For hundreds of gigabytes of climate tensors, an unoptimized shuffle triggers severe network saturation, JVM memory pressure, and out-of-memory (OOM) failures.

2) Multi-Resolution Spatio-Temporal Join: Coupling CR2MET (0.05°, daily), ERA5-Land (0.1°, hourly), and CMIP6 (0.25°, multi-model daily) requires continuous bilinear regridding and psychrometric derivation of relative humidity from dewpoint temperature (T_d) via the Magnus-Tetens formula. This spatial join expands the evaluation domain to $> 7.7 \times 10^9$ hourly state updates for historical reanalysis and $> 1.5 \times 10^{10}$ projections across the 35 CMIP6 GCMs.

3) Stateful Checkpointing and Fault Tolerance: Because the Drought Code has a 52-day memory and accumulates deficits across multi-year droughts, worker failures during long-running batch iterations could cause catastrophic recomputation of RDD lineages. The engine must decouple long time-series into bounded epochs with distributed state checkpointing.

II. Related Work

We classify the state-of-the-art into three distinct research pillars, totaling 18 verified references evaluated according to the criteria defined in [25].

A. Distributed Big Data Systems and Columnar Formats

The foundation of modern distributed data processing originated with Dean and Ghemawat's **MapReduce** [9]. While MapReduce introduced resilient large-scale parallel primitives, its mandatory materialization of intermediate states to disk after every map and reduce phase renders multi-decadal recursive scans computationally intractable.

To overcome disk I/O bottlenecks, Zaharia et al. developed **Apache Spark** and Resilient Distributed Datasets (RDDs) [10], introducing in-memory fault-tolerant abstractions. Spark enables fine-grained control over partitioning through custom hash and range partitioners, allowing narrow lineage dependencies that avoid network shuffles.

Building upon RDDs, Armbrust et al. introduced **Spark SQL** [11], which incorporates the Catalyst query optimizer and the Tungsten execution engine. Catalyst automates logical-to-physical query plan transformations, pushdown predicates, and columnar projection, significantly reducing serialization overhead between the JVM and native memory. For streaming workloads, Armbrust et al. presented **Structured Streaming** [12], providing declarative stateful APIs (`mapGroupsWithState`) with end-to-end exactly-once guarantees.

In parallel, Carbone et al. investigated state management within **Apache Flink** [13], demonstrating that continuous event-driven stream processing with asynchronous Chandy-Lamport state snapshotting can achieve sub-second stateful updates.

Complementing execution engines, Melnik et al. developed **Dremel** [14], introducing nested columnar storage schemes that served as the architectural blueprint for **Apache Parquet**. Parquet utilizes dictionary encoding, bit-packing, and run-length encoding (RLE) to deliver high compression ratios alongside embedded min/max block statistics for efficient data skipping (*predicate pushdown*).

B. Scientific Array Computing and Geospatial Engines

The adaptation of big data frameworks to multi-dimensional scientific arrays has been widely explored. Palamuttam et al. introduced **SciSpark** [15], extending Spark with a specialized array abstraction (`sciRDD`) to perform distributed weather event detection from NetCDF files. However, SciSpark's partitioner divides data across spatial 2D slices at fixed timestamps, which fails to support longitudinal, time-recursive scans without continuous network re-partitioning.

Hu et al. developed **ClimateSpark** [16], an in-memory distributed computing framework tailored to big climate data analytics. ClimateSpark integrates a spatiotemporal index over HDFS blocks to optimize reading NetCDF chunks. Similarly, Li et al. [17] demonstrated spatiotemporal indexing structures on MapReduce to reduce read amplification in MERRA reanalysis datasets. Nonetheless, both frameworks target commutative and associative spatio-temporal reductions (such as global monthly means or regional variance) rather than non-commutative stateful prefix scans.

For vector and raster geospatial operations, Yu, Zhang, and Sarwat introduced **GeoSpark** (now **Apache Sedona**) [18], which embeds spatial indexing (R-tree and Quad-tree) and spatial partitioning into Spark. GeoSpark achieves scalable spatial joins between point sets and polygons, which is critical for intersecting gridded climate rasters with administrative commune boundaries and active fire detections.

On the public cloud, Gorelick et al. established **Google Earth Engine (GEE)** [19], a planetary-scale platform for earth observation data. While GEE provides seamless spatial map algebra over multi-petabyte satellite archives, its computational model is fundamentally optimized for stateless, per-pixel 2D image operations. Expressing multi-decadal recursive scans in GEE via `ee.ImageCollection.iterate()` rapidly exceeds platform recursion limits and user memory budgets, causing pipeline aborts.

Within the scientific Python ecosystem, Rocklin presented **Dask** [20], which, combined with Hoyer and Hamman's **xarray** [21], forms the foundation of the **Pangeo** initiative. Abernathey et al. [22] synthesized the requirements for cloud-native scientific repositories, identifying the fundamental tension between time-chunked and space-chunked layouts in **Zarr** and NetCDF storage. They developed *Rechunker*, a distributed library that coordinates intermediate, two-phase array transpositions in object stores.

C. Fire Weather Climatology and Chile Wildfire Dynamics

The physical underpinnings of the Canadian FWI system were formalized by Van Wagner [4]. Recently, the Canadian Forest Service Fire Danger Group released the **FWI2025** update [23], which introduces the *Hourly Fine Fuel Moisture Code* (Hourly FFMC) to capture diurnal drying cycles, overnight moisture recovery, and shifting peak burning windows.

At the global scale, Vitolo et al. [24] established the Global ECMWF Fire Forecasting (GEFF) reanalysis dataset using ERA5 at a 0.25° ($\sim$28 km) spatial resolution. Similarly, Quilcaille et al. [8] computed global FWI projections under CMIP6 scenarios from 1850 to 2100 at coarse grid scales (1°–2°).

In Chile, Garreaud et al. [3] characterized the dynamical atmospheric drivers of the Central Chile megadrought, demonstrating an ongoing 30% regional rainfall deficit. Boisier [6] produced the CR2MET v2.5 high-resolution precipitation and temperature grids. Cordero et al. [1] and Bowman et al. [2] proved that extreme fire weather driven by ENSO and anthropogenic warming tripled the burned area in Central Chile over the last decade, underscoring the urgent necessity of high-resolution, continuous fire hazard tracking.

III. Discussion of Related Work and Research Gap

Table I provides a systematic taxonomic comparison across candidate distributed processing frameworks.

An inspection of the literature reveals four clear, unaddressed research gaps at the intersection of Big Data Systems and Physical Environmental Modeling:

1) **The Reductive Bias of Big Climate Systems:** Frameworks such as ClimateSpark [16], SciHadoop [17], and standard Spark SQL pipelines are engineered under the assumption that climate analytics consist of *associative and commutative aggregations* (e.g., temporal averages, regional bounding box sums, or climate anomaly indexing). These operations scale seamlessly via tree reductions ($O(\log T)$). In sharp contrast, fire risk modeling requires an inherently non-commutative, non-associative prefix recurrence (Eq. 1) that forces a sequential temporal scan over decades.

TABLE I
TAXONOMIC COMPARISON OF DISTRIBUTED PROCESSING SYSTEMS ON MULTI-DECADAL GRIDDED CLIMATE WORKLOADS

System / Framework	Data Model	Primary Storage Layout	Stateful Temporal Recursion	Shuffle Overhead on Corner-Turn	Primary Research Gap vs. Our Approach
MapReduce [9]	Key-Value pairs	Flat HDFS files (row-oriented)	No native support (disk-bound loops)	Extreme (disk-spill on all map-reduce barriers)	Prohibitive I/O latency when chaining $> 16,000$ sequential daily timesteps.
SciSpark [15]	`sciRDD` multi-dim arrays	NetCDF on HDFS (spatial 2D slices)	Not supported (stateless transformations)	High (requires repartitioning across time)	Partitions by timestamp; cannot execute time-continuous cell scans without global shuffle.
ClimateSpark [16]	Spatiotemporal indexed RDDs	Chunked NetCDF on HDFS	Associative reductions only (means/variances)	Moderate (optimized for spatiotemporal box queries)	Designed for aggregation queries; lacks stateful iterator primitives for recursive moisture accumulation.
Apache Sedona [18]	Spatial RDD / Geometry	Vector GeoJSON, Shapefiles, WKT	Stateless vector spatial joins	Low for vector joins; unadapted for raw 3D arrays	Excellent for polygon/point joins (FIRMS/CONAF), but not designed for 3D tensor time-series rechunking.
Google Earth Engine [19]	Planetary-scale Image Collections	Cloud-optimized 2D tiles (spatial-major)	Severely constrained (`iterate` depth limits)	Hidden (server-side tile computation)	Cannot execute 44-year recursive loops due to memory quotas and iteration recursion stack limits.
Dask + Xarray [20], [22]	N-dimensional blocked arrays	Zarr / NetCDF on Object Storage (S3)	Manual (via `map_blocks` over pre-chunked time)	High without specialized two-phase Rechunker	Requires explicit external rechunking pipeline; lacks built-in query optimizer and streaming state management.
GEFF / CEMS Fire [24]	Fortran monolithic binary pipeline	Coarse GRIB/NetCDF (0.25°)	Sequential in-memory execution per gridpoint	None (runs on single HPC batch nodes)	Monolithic, non-distributed pipeline; spatial resolution ($0.25^\circ \approx 28$ km) misses Chilean mountain valleys.
Our Proposed Architecture	**Spatial-First Partitioned DataFrames**	**Columnar Apache Parquet (ZSTD)**	**Native stateful prefix scan (Arrow + Numba JIT)**	**Zero-shuffle during scan phase** (single one-time repartition)	**Fills the gap:** Combines multi-source spatial fusion, one-time corner-turn, zero-shuffle stateful scans, and streaming updates.

2) **The Storage Layout / Computation Mismatch:** Existing global datasets (such as GEFF [24] and CMIP6-FWI [8]) circumvent distributed systems by running coarse monolithic Fortran/C codes on high-memory HPC nodes. Conversely, cloud-native earth observation engines (like Google Earth Engine [19]) optimize for 2D spatial map tiles at individual timestamps. When scaled to high spatial resolution (0.05°) over 44–75 years (1.6×10^4 to 6.5×10^5 timesteps), neither paradigm scales: HPC nodes run out of memory, while GEE exhausts recursion execution stacks.
3) **The Multi-Source Fusion Void:** No existing system reconciles high-resolution topographic precipitation/temperature grids with hourly atmospheric reanalysis and global multi-model projection ensembles in a unified distributed execution graph.
4) **The Temporal Parallelism and Associative Scan Dilemma:** In a dedicated architectural study led by co-author B. Zúñiga, we investigated whether the temporal axis itself could be parallelized via parallel prefix scans [26]. In linear recurrent systems and state-space sequence models, associative scan primitives convert sequential chains into $O(\log T)$ depth parallel trees [27], [28]. However, the FWI transition operator Φ (Eq. 1) comprises non-linear piecewise branches (separate rain saturation logarithmic curves vs. exponential drying) that lack an exact closed-form associative composition. Nevertheless, because fuel moisture exhibits finite physical memory ($\tau \approx 16$ h for FFMC, $\sim$12 d for DMC, and $\sim$52 d for DC [4], [29]), Zúñiga formulated the **Temporal Warm-Up Halo Hypothesis**: partitioning multi-decadal series into bounded temporal blocks with an overlapping warm-up buffer of 90–180 days. This bounds numerical deviation below 0.1% relative

TABLE II
INTEGRATED DATA ECOSYSTEM AND STORAGE SIZING

Dataset	Variables	Grid / Freq.	Volume
CR2MET v2.5	`pr, tmin, tmax`	0.05° / Day	9.3 GB
ERA5-Land	`t2m, d2m, u, v, tp`	0.10° / Hour	165 GB
NEX-GDDP	`tas, pr, hurs, sfcW`	0.25° / CMIP6	420 GB
CEMS FWI	Historical `FWI, FFMC, DC`	0.25° / Day	12 GB
NASA FIRMS	Active fire (`frp`, pts)	375 m / Daily	450 MB
CONAF Cat.	Ignition logs & area	Point / Event	85 MB

to the exact sequential trajectory, unlocking potential 2D spatio-temporal parallelization for massive CMIP6 ensembles.

Our work directly targets these gaps by formulating a **Two-Phase Architecture: One-Time Distributed Corner-Turn followed by Zero-Shuffle Vectorized Stateful Processing**.

IV. PROPOSED SOLUTION ARCHITECTURE

A. Data Ecosystem and Volume Specifications

The architecture integrates six verified heterogeneous data sources, summarized in Table II. The aggregated historical and projection volume spans hundreds of gigabytes of compressed tensors, expanding to > 2 TB uncompressed, validating the project as a genuine Big Data processing scenario.

B. Four-Stage Distributed Workflow

The workflow is divided into four modular processing stages:

1) Stage 1: Multi-Source Distributed Ingestion and Fusion: Worker nodes read spatial partitions of CR2MET NetCDF files alongside ERA5-Land reanalysis from object storage. For each hourly timestamp in ERA5-Land, saturation vapor pressure (e_s) and actual vapor pressure (e) are computed from T and T_d using the Magnus-Tetens relationship:

$$e(T) = 6.112 \cdot \exp\left(\frac{17.67 \cdot T}{T + 243.5}\right) \quad \text{[hPa]} \tag{4}$$

$$RH = \min\left(100.0,\, 100.0 \cdot \frac{e(T_d)}{e(T)}\right) \quad [\%] \tag{5}$$

Wind speed magnitude is derived as $W = 3.6\sqrt{u_{10}^2 + v_{10}^2}$ km/h.

Furthermore, to reconcile the temporal sampling discrepancy between daily CR2MET extremes and the standard CFFDRS noon observation protocol, we adopt the diurnal adjustment scheme (S_2) formulated by co-author B. Zúñiga: ERA5-Land is sampled at 16:00 UTC ($\approx$ 12:00 Local Standard Time, UTC-4), and local temperature is bias-adjusted against CR2MET topography:

$$T_{\text{noon}} = T_{\text{ERA5, 16:00}} + (T_{\max,\text{CR2MET}} - T_{\max,\text{ERA5}}) \tag{6}$$

This couples CR2MET's high-resolution orographic temperature gradient with ERA5-Land's psychrometric dewpoint, mitigating Andean cold biases. Using Apache Spark DataFrames, values are bilinearly interpolated onto CR2MET's $0.05°$ grid and indexed via the **Uber H3 Hexagonal Spatial Index** (resolution 6, $\sim$3.2 km aperture).

2) Stage 2: Distributed Corner-Turn and Columnar Materialization: To permanently resolve the time-major read amplification bottleneck, Spark executes a controlled global partition:

```
df.repartitionByRange(P, "h3_id")
  .sortWithinPartitions("h3_id", "time")
```

where P is the cluster parallelism factor. The transposed data is materialized into **Apache Parquet files compressed with Zstandard (ZSTD)**, partitioned by `h3_id`. Each Parquet file encapsulates the entire contiguous multi-decadal time-series for a local spatial cluster of cells. Embedded dictionary encoding and min/max block statistics eliminate redundant metadata reads.

3) Stage 3: Zero-Shuffle Vectorized Stateful Scan: Because each executor now possesses the complete temporal trajectory for its assigned spatial partition, FWI evaluation proceeds with **zero inter-node network communication**:

```
df.groupBy("h3_id")
  .applyInPandas(compute_fwi_numba, schema)
```

Inside each executor, PyArrow streams raw memory chunks to a **Numba JIT-compiled C-kernel** that computes the recursive state transitions for both daily Van Wagner (1987) and hourly FWI2025 algorithms. The terminal state vector $\mathcal{S}_{c,T}$ is persisted to a lightweight checkpoint table, while the computed FWI time-series are stored in Snappy-compressed columnar Parquet tables.

4) Stage 4: Stateful Incremental Streaming and Spatial Validation: When near-real-time observations arrive via the DMC API or operational forecasts, **Spark Structured Streaming** ingests the incremental batch, loads $\mathcal{S}_{c,T}$ from checkpoint storage, and updates the hazard surface in sub-second latency using `mapGroupsWithState`. Concurrently, **Apache Sedona** performs a distributed spatial join between high-FWI cells ($FWI > 50$) and NASA FIRMS active fire pixels / CONAF historical ignition logs to evaluate predictive skill.

C. Simulation Methodology and Containerized Cluster Design

To achieve rigorous, reproducible evaluation within student computational constraints, simulations will execute on an isolated **Docker-based cluster environment**:

- **Topology:** 1 Spark Master container and $N \in \{1, 2, 4, 8\}$ Spark Worker containers, each constrained via Docker `cgroups` to strict CPU and RAM limits (e.g., 1 vCPU, 2 GB RAM per worker).
- **Storage Layer:** A local **MinIO** container providing S3-compliant distributed object storage, enabling realistic measurement of network socket I/O throughput.
- **Baseline Cluster:** A mirrored containerized **Dask/Xarray + Zarr** cluster subjected to identical resource caps.

D. Experimental Evaluation Plan and COST Benchmarking

To rigorously answer whether distributed scale-out is justified over optimized scale-up execution, our evaluation framework—designed by co-author B. Zúñiga—establishes a four-tier baseline hierarchy:

- B_0 **(Naive Single-Cell Sequential):** A baseline Python script evaluating cells sequentially in pure interpreter mode (measuring the unoptimized worst case).
- B_1 **(Vectorized Single-Node Ceiling):** An in-memory multi-threaded engine using `xarray` and `Numba JIT` compiled C-loops (looping solely over time with array-packed cells), serving as the high-performance single-node ceiling.
- B_2 **(Dask/Zarr Baseline Cluster):** A containerized Dask cluster utilizing chunked Zarr arrays over MinIO object storage.
- S **(Proposed Architecture):** Our Spatial-First Apache Spark engine over Parquet (ZSTD) with Arrow-Numba stateful kernel.

The forthcoming experimental campaign will measure five core systems metrics:

1) **Configuration that Outperforms a Single Thread (COST):** Following McSherry et al. [30], we report the empirical COST metric: the exact number of Spark cluster cores required before distributed execution outperforms the highly optimized single-node vectorized baseline B_1. This directly answers skepticism regarding framework overheads and identifies the volume threshold where Spark becomes necessary.
2) **Bottleneck Analysis and Blocked Time Breakdown:** Following Ousterhout et al. [31], we decompose worker execution time into CPU computation, JVM garbage collection, disk I/O, and network serialization to verify that spatial-first Parquet eliminates network communication during the stateful scan.
3) **Strong Scaling and Speedup:** Measuring total runtime $T(N)$ across $N \in \{1, 2, 4, 8\}$ workers for fixed decadal spans, evaluating empirical Speedup $S(N) = T(1)/T(N)$ against Amdahl's Law.
4) **Shuffle Traffic and Read Amplification:** Monitoring Spark metrics (`ShuffleWriteBytes`, `SpillToDiskBytes`) to verify that spatial-first Parquet eliminates shuffle compared to naive time partitioning.
5) **Domain Calibration and ROC-AUC:** Assessing Receiver Operating Characteristic curves of calculated FWI against CONAF ignition coordinates during the 2017 and 2023 disaster windows.

V. Conclusion

This paper establishes the state-of-the-art and technical solution architecture for distributed fire weather danger estimation across Chile. By revealing the data shortfalls of CR2MET v2.5 and formalizing the non-commutative temporal recurrence of the Canadian FWI, we demonstrated that the core computational bottleneck lies in the distributed corner-turn shuffle and multi-source spatial fusion. Our proposed Apache Spark and Parquet architecture with spatial-first partitioning eliminates network shuffling during recursive evaluation, providing a scalable, scientifically rigorous baseline for national wildfire resilience.

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